

A sparse penalty whose proximal map is not soft thresholding

Disproof of Conjecture 00000008855

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Penalty and proximal definition

The source asserts that proximal operators of sparse penalties are always soft thresholding, without restricting the penalty to a convex class. Consider the standard scalar nonzero-count penalty

$$P(y) = \|y\|_0 = \begin{cases} 0 & y = 0, \\ 1 & y \neq 0. \end{cases}$$

This is a sparsity penalty: it charges one for a nonzero coordinate and zero for a zero coordinate. With unit proximal parameter, define the proximal graph by global minimization of

$$F_x(y) = \frac{1}{2}(y - x)^2 + P(y), \quad \text{prox}_P(x) = \underset{y \in \mathbb{R}}{\operatorname{argmin}} F_x(y).$$

It may be a set in general, but the two inputs below have unique minimizers.

Two exact global minima

At $x = 2$, $F_2(2) = 1$ whereas $F_2(0) = 2$. For every nonzero y , $F_2(y) = 1 + (y - 2)^2/2 \geq 1$, with equality if and only if $y = 2$. Thus $\text{prox}_P(2) = \{2\}$. At $x = 1$, $F_1(0) = 1/2$. For every nonzero y , $F_1(y) = 1 + (y - 1)^2/2 \geq 1 > 1/2$. Consequently $\text{prox}_P(1) = \{0\}$. These comparisons check every real competitor; they do not rely on an assumed hard-thresholding table.

No fixed soft threshold can realize this map

For a threshold $\tau \geq 0$, the standard scalar soft threshold is

$$S_\tau(x) = \begin{cases} x - \tau & x > \tau, \\ x + \tau & x < -\tau, \\ 0 & |x| \leq \tau. \end{cases}$$

If S_τ were a proximal map for this penalty, uniqueness at $x = 2$ would require $S_\tau(2) = 2$. When $\tau \geq 2$ its value is zero; when $0 \leq \tau < 2$ its value is $2 - \tau$. The required equality therefore forces $\tau = 0$. But $S_0(1) = 1$, contradicting the unique proximal value zero at $x = 1$. Thus no fixed nonnegative threshold can represent the proximal operator. In particular the customary unit threshold gives $S_1(2) = 1$, which is not a minimizer.

This refutes the first clause of the original conjunction; the separate splitting-rate clause is not needed for the disproof. Soft thresholding for the absolute-value penalty is a different result and does not imply the universal statement for sparse penalties.

Formal verification

Lean defines the actual piecewise nonzero-count penalty, the real quadratic objective, and the proximal relation as comparison with every real competitor. It proves both exact proximal-set characterizations, defines the piecewise soft-threshold function, and proves the impossibility of a threshold valid for every input. The final theorem `conjecture_8855_false` is stronger than a failure at one prescribed threshold: even changing the fixed threshold cannot fix this penalty.